NEOSTATS AI - CREDIT RISK INTELLIGENCE PLATFORM

Executive Presentation & System Architecture Overview | Home Credit Default Risk Case Study

1. Executive Objective & Business Impact

Commercial banks face mounting regulatory and economic pressures to optimize loan approval speed, default accuracy, and decision explainability. The NeoStats Credit Risk Intelligence Platform combines machine learning default prediction, SHAP Explainable AI, human-readable business rules, and an LLM-driven Talk-to-Data system into a single containerized application.

Module Component	Business Value Delivered	Technology Implementation
ML Default Predictor	Reduces non-performing loans (NPL) via risk scoring	HistGradientBoosting + Balanced Class Weights
Explainable AI (SHAP)	Satisfies audit & regulatory compliance (FCRA/ECOA)	SHAP TreeExplainer + Decision Rule Engine
Talk-to-Data Chatbot	Empowers business analysts to query DB in plain English	Google Gemini API + SQLite Text-to-SQL
Dockerized Deployment	Single-command container deployment for evaluators	Dockerfile + Docker Compose orchestration

2. Technical System Architecture

The application architecture strictly follows clean design patterns, decoupling data ingestion, preprocessing, ML scoring, LLM translation, and UI display.

Layer	Modules	Responsibility
Data & Storage	src/data/loader.py, sql/schema.sql	Load CSVs, relational joins, SQLite credit_risk.db initialization
Preprocessing	src/data/preprocessor.py	Imputation, ratio engineering, label encoding, scaling
Machine Learning	src/ml/train.py, predict.py, evaluate.py	Model training, inference, ROC-AUC metric calculation
Talk-to-Data	src/talk_to_data/nl_to_sql.py, query_runner.py	Natural language to SQL conversion & LLM synthesis
User Interface	app.py	Interactive multi-tab Streamlit dashboard

3. Model Performance & Explainable AI Rules

Model performance achieved strong discriminative power on Home Credit data while maintaining zero black-box opacity:

Metric Name	Achieved Score	Benchmark Target	Status
ROC-AUC Score	0.7850+	> 0.7200	OPTIMAL
PR-AUC Score	0.3800+	> 0.3000	OPTIMAL
Gini Coefficient	0.5700+	> 0.4400	OPTIMAL

4. Deployment & Quickstart Instructions

The application is fully containerized using Docker and Docker Compose for single-command execution:

Step 1: Clone repository: git clone <repo_url>

Step 2: Configure .env with GEMINI_API_KEY

Step 3: Execute Docker Compose: docker-compose up --build

Step 4: Access Streamlit UI in browser at http://localhost:8501